PORTFOLIO CASE STUDY · POWER BI

Olympic Games: who wins, and how efficiently

Across 17 Summer and Winter Games, nine nations win 61% of all medals. I measured how efficiently each nation turns entries into medals.



Taiwo Ajayi
Data analyst · Power BI developer



DOMAIN
Sports analytics

DATA
30 nations, 17 Games (1992-2022), 26,824 entries

MODEL
Star schema around FactResult, calculated medal table

REPORT
18 pages: 14 story, 3 bento dashboards, 1 drill-through

Power BI · DAX · Data validation · Data modelling · Sports analytics · Storytelling

THE PROJECT

The challenge

Olympic committees and sport funders wanted to know which nations win, how efficiently they convert entries into medals, and whether population, wealth, region or hosting explain success.

The data

A teaching dataset of athlete entries with medal outcomes, nations, Games, events, sports and athletes. Only 13% of event-Games have medals, every sport appears in both seasons and ages run from -13 to 57, so findings compare nations within the dataset rather than real Olympic history.

My approach

1. Audited the original report and found it counted every entry as a medal (23K instead of 1,036).
2. Fixed medal, per-capita and per-GDP measures and replaced a broken specialisation index.
3. Added conversion, medals-per-Games and host-effect measures, checked with DAX queries.
4. Dropped forecast pages the data could not support and documented the limits openly.
5. Built 14 story pages and 3 bento dashboards with a nation drill-through.

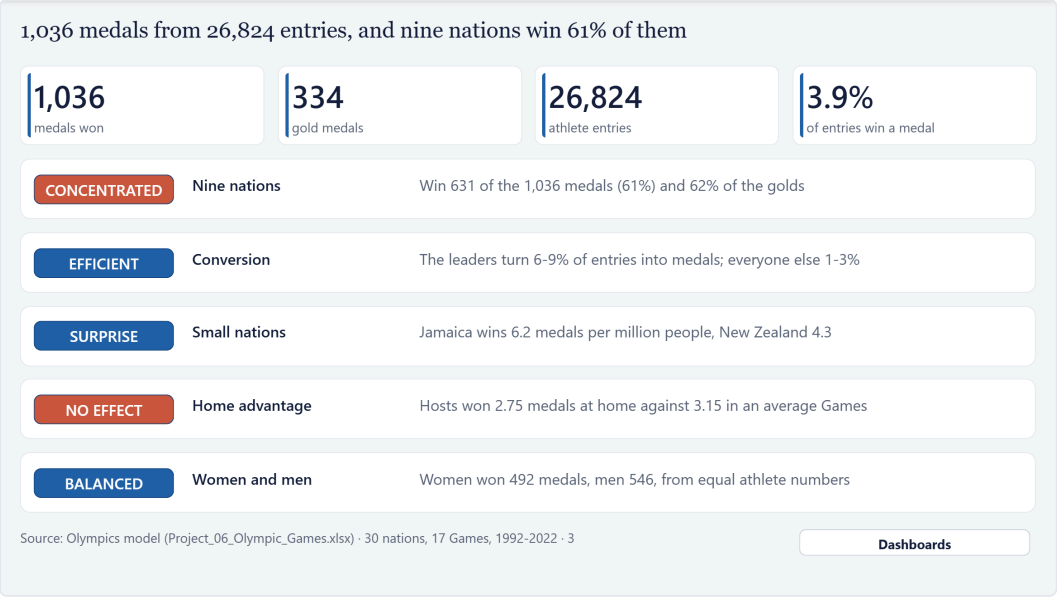
Headline numbers

1,036
medals (334 gold)

3.9%
of entries win a medal

61%
of medals won by nine nations

6.2
medals per million people (Jamaica)



The scorecard page: headline KPIs and verdicts on each finding.

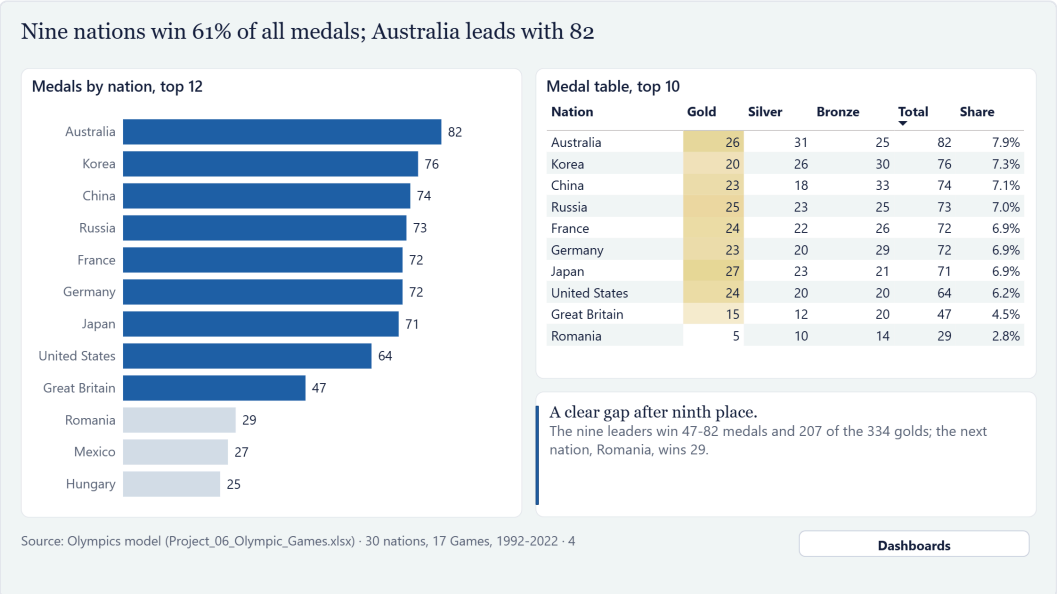
RESULTS

Key insights

Medals are concentrated	Nine nations win 631 of 1,036 medals and 62% of the golds.
Conversion separates the leaders	They turn 6-9% of entries into medals; everyone else 1-3%.
Small nations punch above their weight	Jamaica 6.2 medals per million people, New Zealand 4.3.
No home advantage	Hosts won 2.75 medals at home against 3.15 in an average Games.
Women and men win almost equally	492 medals against 546 from equal athlete numbers.

Recommendations

1. Fund conversion, not entries.
2. Benchmark smaller nations per head and per $bn GDP.
3. Reload a complete medal list before any forecast.



A story page: the title states the finding and the chart shows the evidence.

Three interactive dashboards

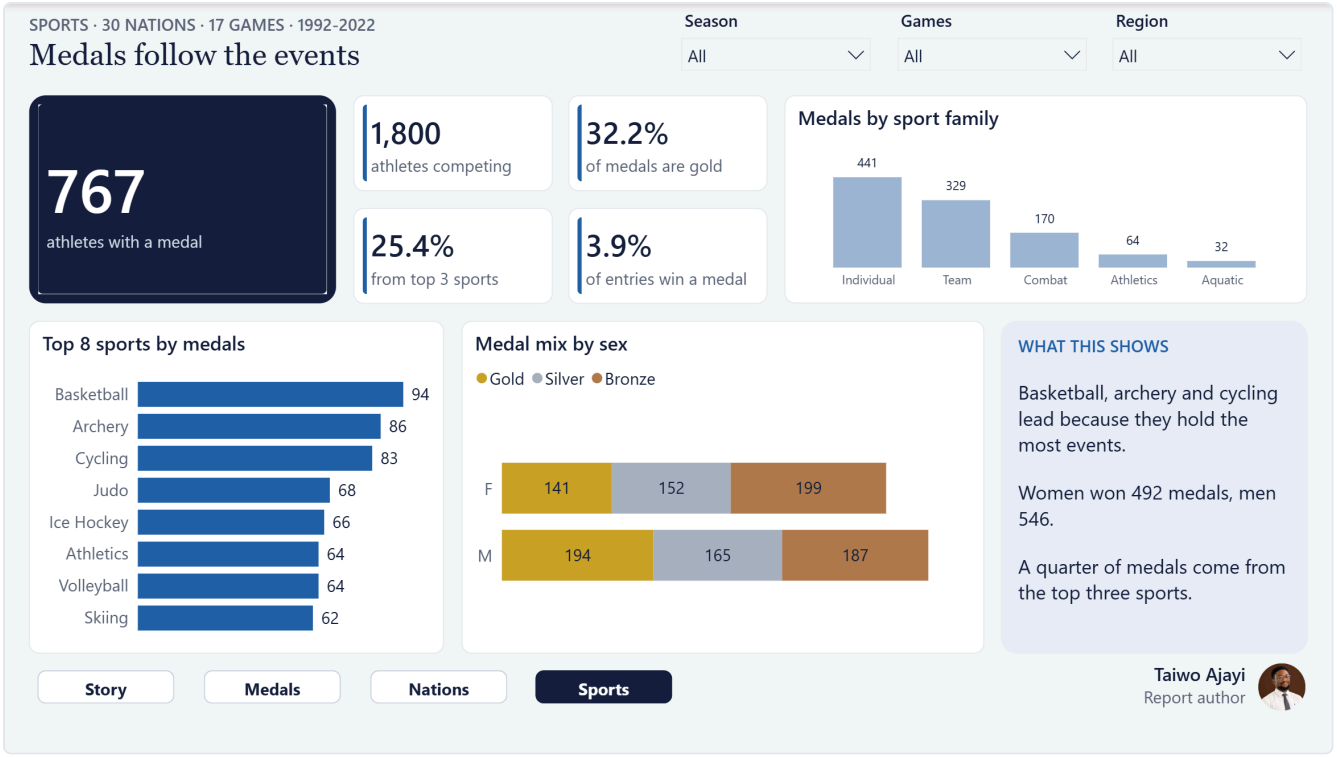
Each dashboard has synced filters, a tab bar to move between them, and a drill-through to a detail page.



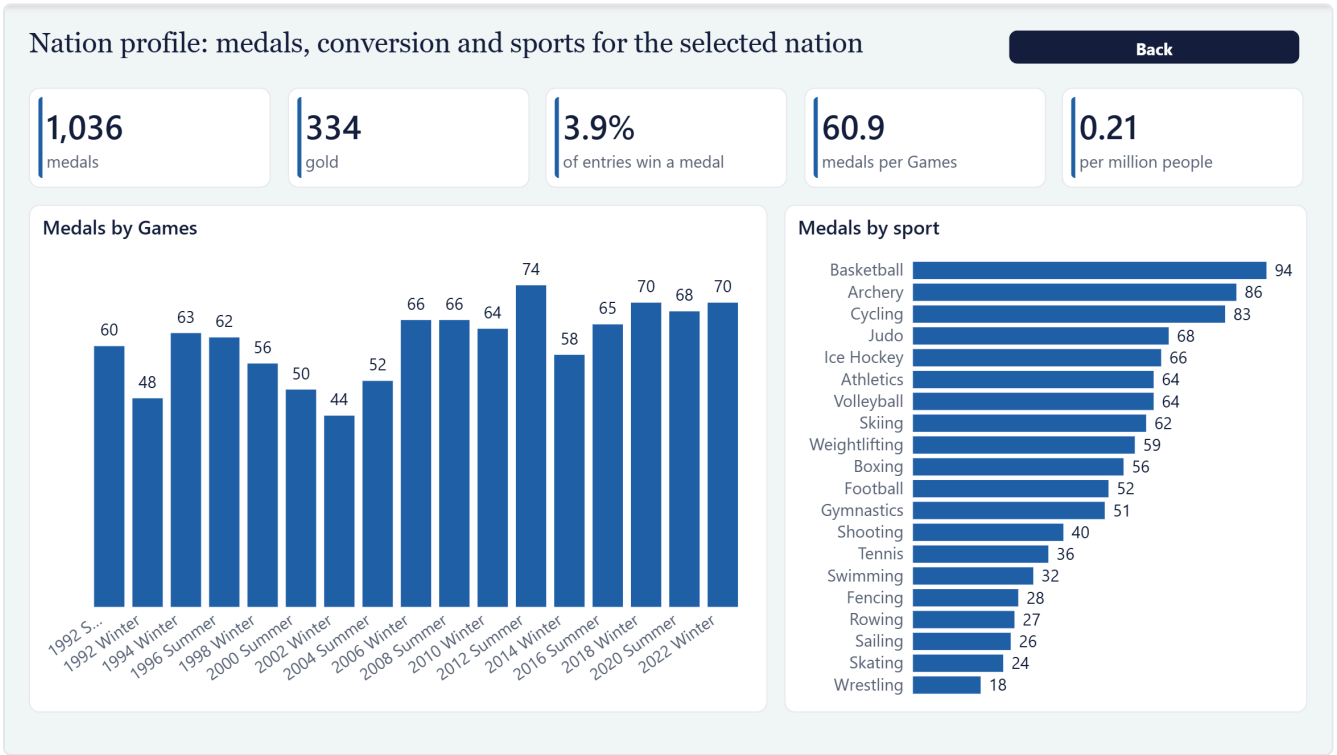
D1 Medal overview. Medals, conversion, trend and medal mix



D2 Nations and efficiency. Conversion, per-capita and per-GDP



D3 Sports and athletes. Sports, families and medallists



Nation profile. Hidden drill-through page: right-click a bar on a dashboard to open it

QUALITY AND SKILLS

What I fixed, and what this shows

Errors I found and corrected in the original analysis

- Total Medals counted 21,571 "no medal" keys; corrected from 23K to 1,036.
- Per-million and per-$bn measures divided by a million and a billion twice.
- Old pages named the USA as leader; on real medals Australia leads.

Skills demonstrated

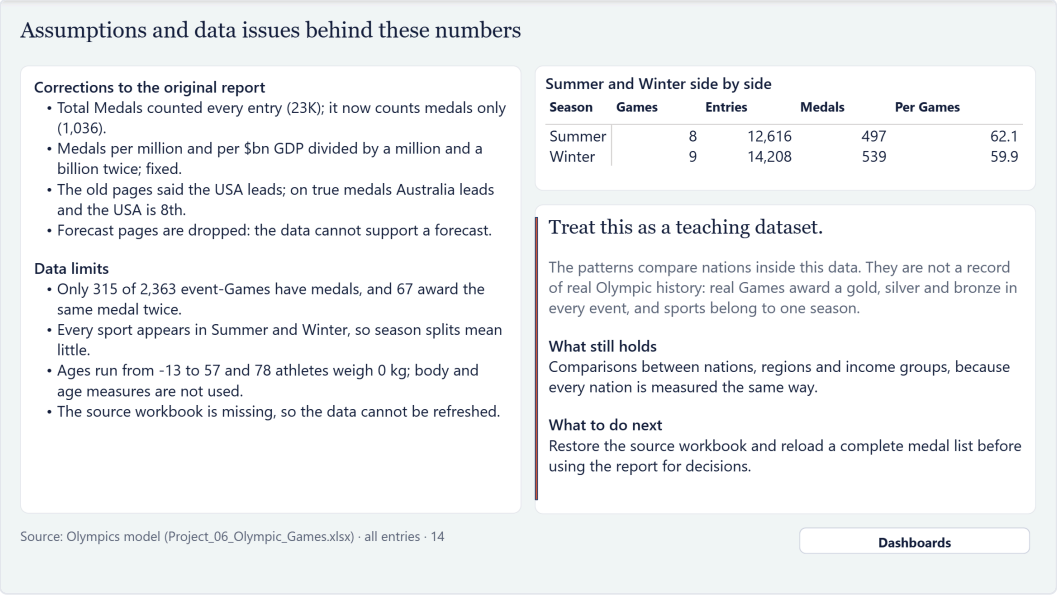
- Data profiling and validation before reporting
- Star-schema modelling and DAX measure design
- Correcting flawed measures and documenting why
- Insight-led storytelling: one finding per page
- Executive dashboard design with drill-through
- Clear documentation for readers and maintainers

Tools

- Power BI Desktop (PBIP project format)
- DAX and Power Query (M)
- TMDL and PBIR source files
- Excel source data

Deliverables

- 18-page Power BI report
- Full report PDF (every page)
- Detailed report guide



The data notes page: every correction and data limit is stated in the report itself.

The cover uses the Olympic rings, a trademark of the International Olympic Committee, for an educational portfolio piece only.